

2.6.38

AI25BTECH11027 - NAGA BHUVANA

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Question:

If $\mathbf{a} = \hat{i} + \hat{j} + \hat{k}$ and $\mathbf{b} = \hat{j} - \hat{k}$, find a vector $\mathbf{c}$ such that $\mathbf{a} \times \mathbf{c} = \mathbf{b}$ and $\mathbf{a} \cdot \mathbf{c} = 3$

Solution:

$$\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad (0.1)$$

$$\mathbf{a}^T \mathbf{c} = 3 \quad (0.2)$$

$$\mathbf{b}^T \mathbf{c} = 0 \quad (0.3)$$

$$(\mathbf{a} \quad \mathbf{b})^T \mathbf{c} = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad (0.4)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \mathbf{c} = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad (0.5)$$

By solving

$$\mathbf{c} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 - 2\lambda \\ \lambda \\ \lambda \end{pmatrix} \quad (0.6)$$

Now consider Scalar triple product of $\mathbf{a}, \mathbf{c}$ and $\mathbf{b}$

$$(\mathbf{a} \times \mathbf{c}) \cdot \mathbf{b} = \|\mathbf{b}\|^2 = 2 \quad (0.7)$$

$$(\mathbf{a} \times \mathbf{c}) \cdot \mathbf{b} = \det \begin{pmatrix} 1 & 3 - 2\lambda & 0 \\ 1 & \lambda & 1 \\ 1 & \lambda & -1 \end{pmatrix} = 2 \quad (0.8)$$

$$6 - 6\lambda = 2 \quad (0.9)$$

$$\lambda = \frac{2}{3} \quad (0.10)$$

$$\mathbf{c} = \begin{pmatrix} 5 \\ 3 \\ 3 \\ 3 \\ 3 \end{pmatrix} \quad (0.11)$$

$$\therefore \mathbf{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k} \quad (0.12)$$

Graphical Representation

Vectors a , b , and c

